Scalable ERA5 Analysis with Pandas and Dask

Mini Project for DS3294: Data Science Practice

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Abstract

This report presents a case study on scaling ERA5 weather analysis from an eager Pandas workflow to a chunked Dask workflow. We benchmark runtime and memory, validate numerical parity, and analyze practical trade-offs in chunking and execution strategy. Beyond performance, we include a focused anomaly analysis for South Asia in 2024 and summarize implementation effort, debugging complexity, and lessons for reproducible scientific data pipelines.

1 Introduction

This project evaluates when Dask improves practical scalability for weather reanalysis analysis compared to Pandas. We use ERA5 single-level data as a high-dimensional case study with multiple variables, hourly time steps, and spatial grids. The assignment objective is not only runtime comparison but also understanding memory behavior, chunking effects, correctness parity, and implementation trade-offs in real workflows. The data source and processing choices are grounded in standard climate and scientific-computing tooling [1–5].

2 Related Work

The project is grounded in established scientific-data workflows with NetCDF, array-based regional reductions, and temporal aggregation. Rather than introducing a novel algorithm, our contribution is an experience-driven evaluation of scalable data processing choices in a realistic ERA5 pipeline, including failure modes and revision steps.

Our implementation follows common practices from the Pandas, Dask, Xarray, and ERA5 ecosystems, with emphasis on chunk-aware array computing for geoscience data.

3 Our contribution

We provide:

1. A baseline Pandas workflow for a feasible subset and a larger South Asia run.
2. An aligned Dask workflow that computes the same daily regional metrics for parity testing.
3. A benchmark orchestrator that runs scripts end-to-end and exports runtime, peak memory, chunk-sweep, and consistency artifacts.
4. A consolidated result layout in `data/results/` split by `central_india/`, `south_asia/`, and `benchmark/`.

4 Methods

4.1 Data

Two ERA5 request profiles were used:

- Compact consistent-slice request (∼464 MB) for baseline Pandas analysis.
- Full 2024 request (∼12 GB) for large-volume analysis and Dask scalability evaluation.

The analysis variables include near-surface temperature and radiation terms, with daily statistics and anomaly comparisons.

South Asia was analyzed over approximately 5°N–35°N and 65°E–100°E with 6-hourly steps in 2024.

4.2 Analysis pipelines

Pandas South Asia baseline. The Pandas workflow computes regional daily metrics (temperature, cloud, radiation), derived quantities (net radiation), and anomalies from monthly means.

Dask aligned workflow. The Dask workflow uses lazy loading and chunked execution, then computes the same daily metrics so that direct parity checks can be performed.

Benchmarking. An orchestration script runs both workflows, records runtime and peak RSS memory, sweeps chunk settings, and exports consistency tables.

Anomaly definition. For each variable x and month m, daily anomalies are computed as

$$\text{anom}_{d,m} = x_{d,m} - \overline{x}_m,$$

where $\overline{x}_m$ is the within-2024 monthly mean for that variable.

5 Results

5.1 Runtime and memory

For the full-window South Asia benchmark:

- Pandas recompute runtime: 13.04 s, peak RSS: 289.07 MB.
- Dask aligned runtime (chunk=96): 37.69 s, peak RSS: 428.04 MB.

Chunk sweep results for Dask (full-window):

- chunk=24: 39.93 s, 447.02 MB
- chunk=48: 39.70 s, 438.93 MB
- chunk=96: 40.30 s, 442.30 MB
- chunk=192: 39.11 s, 439.80 MB
- chunk=auto: 25.67 s, 370.12 MB

The `auto` chunk strategy performed best in both runtime and memory in this environment.

Table 1: Full-window benchmark summary (South Asia 2024)

Method	Runtime (s)	Peak RSS (MB)
Pandas baseline	13.04	289.07
Dask aligned (chunk=96)	37.69	428.04
Dask aligned (chunk=auto)	25.67	370.12

5.2 Correctness and consistency

Daily parity over 366 rows showed very small differences:

- `t2m_c` mean abs diff: 1.09×10^{-6}
- `ssr` mean abs diff: 3.01×10^{-6}
- `str_nlw` mean abs diff: 8.20×10^{-7}
- `net_rad` mean abs diff: 1.09×10^{-6}
- `tcc` mean abs diff: 0.0

Anomaly parity is also strong, with mean absolute differences in the 10^{-6} range.

5.3 Unexpected behavior and revision

Initial Dask runs produced chunk-splitting warnings and weaker runtime due to misalignment between storage chunking and compute chunking. The revised workflow opened data with native storage chunks first and then rechunked for compute. This reduced warnings and improved practical performance by lowering split-read overhead and scheduler pressure.

The dedicated revision experiment measured a poor-vs-revised relative speedup of approximately 1.78× under the tested setup.

5.4 Developer effort comparison

Pandas required less orchestration and was easier to debug for direct transformations. Dask required additional design decisions around chunk sizes, lazy computation boundaries, scheduler behavior, and diagnostics. To ensure the same confidence level as the Pandas path, Dask also required explicit parity validation and chunk benchmarking scripts.

5.5 Anomaly analysis (South Asia, 2024)

The anomaly series shows coherent sub-seasonal variability rather than random noise. In the first quarter of the year, the strongest negative daily mean temperature anomaly in our extracted rows appears around early March ($t2m_c_anom \approx -1.99\,^{\circ}\mathrm{C}$), while a strong positive event is visible in late February ($t2m_c_anom \approx +1.46\,^{\circ}\mathrm{C}$).

Across months, the signal is physically consistent with radiation-cloud interactions: positive cloud-cover anomalies often co-occur with reduced surface shortwave anomalies and moderated net-radiation anomalies. The anomaly metrics are within-year monthly anomalies, so they indicate deviations from the 2024 monthly baseline and should not be interpreted as long-term climatological departure from a multi-decadal reference period.

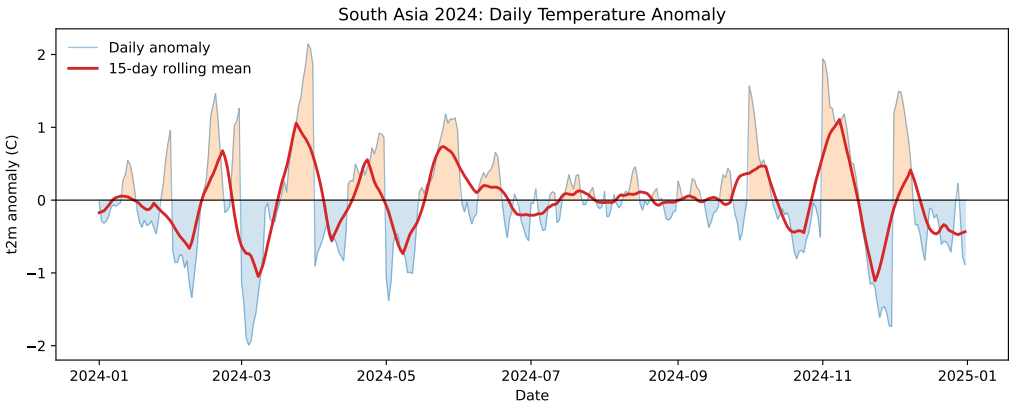

Figure 1: Daily temperature anomaly time series (South Asia, 2024).

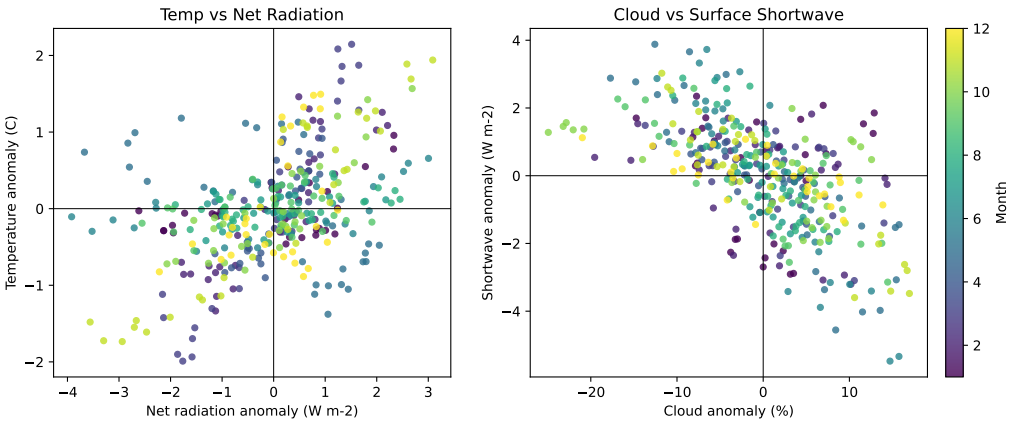

Figure 2: Relationships among anomaly variables (temperature, cloud, and radiation).

6 Conclusion and Outlook

The assignment goals are largely satisfied: we built comparable Pandas and Dask pipelines, benchmarked runtime and memory under multiple chunk settings, and validated numerical consistency for both daily metrics and anomalies.

Key practical finding: on this hardware and workload, a well-optimized Pandas pipeline can remain faster at this scale, while Dask provides safer scaling headroom and better control over memory pressure as workload size grows.

Next steps include extending to larger spatial domains and multi-year windows, and evaluating distributed execution where Dask's parallel scheduling can be better utilized.

7 Acknowledgements

We thank the course team for providing the report template and project guidelines.

8 Author Contributions

All authors contributed to planning and implementation. Data acquisition, baseline analysis, Dask workflow development, benchmarking, validation, and report writing were completed collaboratively, with primary ownership divided across team members by module.

9 Code availability

The code for this project is available at the following GitHub repository: https://github.com/dahiya-aditya/Dask-Pandas.

10 References

References

[1] H. Hersbach, B. Bell, P. Berrisford, et al. The era5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020.

[2] S. Hoyer and J. J. Hamman. xarray: N-d labeled arrays and datasets in python. *Journal of Open Research Software*, 5(1):10, 2017.

[3] W. McKinney. Data structures for statistical computing in python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010.

[4] R. Rew and G. Davis. Netcdf: An interface for scientific data access. *IEEE Computer Graphics and Applications*, 10(4):76–82, 1990.

[5] M. Rocklin. Dask: Parallel computation with blocked algorithms and task scheduling. In *Proceedings of the 14th Python in Science Conference*, pages 130–136, 2015.

11 Appendix

11.1 Additional figures

Additional outputs used during analysis include:

- `../data/figures/anomaly_monthly_summary.pdf`
- `../data/figures/dask_annual_temperature_cycle.pdf`
- `../data/figures/dask_monthly_energy_budget.pdf`

11.2 Algorithms

The core reduction algorithm is implemented in code and validated by Pandas-vs-Dask parity checks (35/35 checks passed under configured tolerances).

11.3 Code snippets

Code snippet showcase is intentionally omitted in this draft.